EXISTENCE OF A K-FLAT RESOLUTION BY FLAT MODULES WITH A TERMWISE SURJECTIVE MAP

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot (Xy-pic setup plus theorem declarations and mathematical macros). Nothing was added, removed or redefined, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

Editorial note. The citation fields (source_id, source_version, authors, source_url, license.name, source_title) are identical to those of every other card this lane has produced from this chapter, so the batch selector treats them as one source.

Allowed standard prerequisites (editorial).

- Existence of the special direct system of bounded-above approximations of a complex by complexes of flat modules. Derived Categories, Lemma \ref{derived-lemma-special-direct-system}, cited at proof lines 14388-14392. Produces the system of complexes and the compatible maps whose colimit is the required resolution.
- A bounded above complex of flat modules is K-flat. Lemma \ref{lemma-derived-tor-quasi-isomorphism}, cited at proof lines 14395-14396. Makes each term of the system K-flat.
- K-flatness is preserved by filtered colimits, and filtered colimits of flat modules are flat. Lemma \ref{lemma-colimit-K-flat} and Algebra, Lemma \ref{algebra-lemma-colimit-flat}, cited at proof lines 14400 and 14403-14405. Carry K-flatness and flatness of the terms to the colimit.

Source reference. Original lemma 06Y4: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 8873–8873. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 06XX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 8866–8866. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 064K: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7869–7869. Navigation pointer to a statement the

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 06Y3: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 8872–8872. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 05UT: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7527–7527. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let R be a ring. For any complex $M^\bullet$ there exists a K-flat complex $K^\bullet$ whose terms are flat R-modules and a quasi-isomorphism $K^\bullet \to M^\bullet$ which is termwise surjective.*

Proof. Let $\mathcal{P} \subset \mathrm{Ob}(\mathrm{Mod}_R)$ be the class of flat R-modules. By Derived Categories, Lemma 06XX there exists a system $K_1^\bullet \to K_2^\bullet \to \ldots$ and a diagram

$$\begin{array}{ccccc} K_1^\bullet & \longrightarrow & K_2^\bullet & \longrightarrow & \ldots \\ \downarrow & & \downarrow & & \\ \tau_{\leq 1}M^\bullet & \longrightarrow & \tau_{\leq 2}M^\bullet & \longrightarrow & \ldots \end{array}$$

with the properties (1), (2), (3) listed in that lemma. These properties imply each complex $K_i^\bullet$ is a bounded above complex of flat modules. Hence $K_i^\bullet$ is K-flat by Lemma 064K. The induced map $\mathrm{colim}_i K_i^\bullet \to M^\bullet$ is a quasi-isomorphism and termwise surjective by construction. The complex $\mathrm{colim}_i K_i^\bullet$ is K-flat by Lemma 06Y3. The terms $\mathrm{colim} K_i^n$ are flat because filtered colimits of flat modules are flat, see Algebra, Lemma 05UT. $\square$